Loanliness: Predicting Loan Repayment Ability with Machine Learning

UE24CS352A Machine Learning Mini-Project, PES University | Kundhan (PES1UG24AM327) and Tarun Gowda (PES1UG24AM303)

1. Problem statement

Lenders decide whether to approve a loan largely from credit history, which shuts out students, young workers and anyone else with a thin credit file. Following Liang, Jin and Wang (Stanford CS229), we frame repayment ability as binary classification: from an applicant's loan application and any past borrowing records, predict whether they will have payment difficulties (TARGET = 1). Because defaulters are rare, we judge models by how well they *rank* risk (ROC-AUC, PR-AUC) and by recall and precision at a chosen cut-off, not by accuracy. We also re-test three claims from the reference paper: that down-sampling gives the best models, that K-means clusters plus one model per cluster improve accuracy, and which features drive repayment.

2. Dataset

Kaggle's *Home Credit Default Risk* data: 307,511 labelled applicants. The main table has 120 attributes (income, loan amount and annuity, age, employment, family, housing, region, documents, and three normalised external credit scores EXT_SOURCE_1-3). Six history tables add 58 million rows of past behaviour: 1.7M loans reported to the credit bureau with 27M monthly statuses, 1.7M previous Home Credit applications, 13.6M instalment payments, and monthly POS/cash and credit-card balances. Only 8.1% of applicants defaulted, so predicting "repays" for everyone already scores 91.9% accuracy. We fixed one data error: DAYS_EMPLOYED (and three date columns in previous applications) use 365,243 days, about 1,000 years, as a placeholder; it covers 18% of applicants, mostly pensioners, and became a missing value plus a flag.

3. Approach

Features. 13 affordability features from the application (credit/income, annuity/income, annuity/credit as repayment speed, goods price/credit, employment/age, income per person, and summaries of the external scores), plus 122 history features: each history table is aggregated to one row per applicant, for example share of instalments paid late (overall, last year, last 6 months), days past due, bureau debt/credit ratio, refused applications and credit-card utilisation. The label is never used in these, so they cannot leak it. 256 features in total. **Protocol.** A stratified 80/20 split holds out 61,503 applicants, touched once at the end. Everything else uses 5-fold stratified cross-validation on the training part. Imputation, scaling, one-hot encoding and any resampling sit inside an imblearn Pipeline and are re-fitted on each training fold, so validation data never leaks into training (the paper does not say how it avoided this). The decision threshold maximises F1 on out-of-fold predictions and is then frozen. **Models.** The paper's five families: logistic regression (class-weighted), Gaussian Naive Bayes, random forest (class-weighted), a 64-32 MLP (under-sampled folds) and LightGBM. **Extra experiments:** four imbalance strategies, the paper's 50/50 evaluation, K-means (k = 2-5 on 20 PCA components) with one LightGBM per cluster, and ablations without history, external scores or gender.

4. Implementation and results

The code is a small Python package: `src/history.py` aggregates the history tables (about one minute), `src/data.py` loads, cleans and builds features, `src/train.py` runs every experiment and writes all numbers to `reports/results.json`, and a Streamlit demo, `app/app.py`, scores an applicant, applies an adjustable threshold and shows the ten features that moved that applicant's score (per-row SHAP values from LightGBM). The full run takes 61 minutes on two CPU cores. Table 1 gives held-out results.

Model	CV ROC-AUC	Test ROC-AUC	Test PR-AUC	Recall	Precision	F1
LightGBM	**0.791 ± 0.002**	**0.794**	**0.297**	**0.44**	**0.29**	**0.35**
Logistic Regression	0.775 ± 0.002	0.778	0.264	0.42	0.28	0.33
Random Forest	0.765 ± 0.002	0.768	0.253	0.42	0.25	0.32
MLP (64-32)	0.765 ± 0.004	0.765	0.241	0.45	0.24	0.31
Naive Bayes	0.586 ± 0.047	0.624	0.111	0.72	0.12	0.20

Table 1. Held-out test set (61,503 applicants). Recall, precision and F1 at the cut-off chosen by cross-validation.

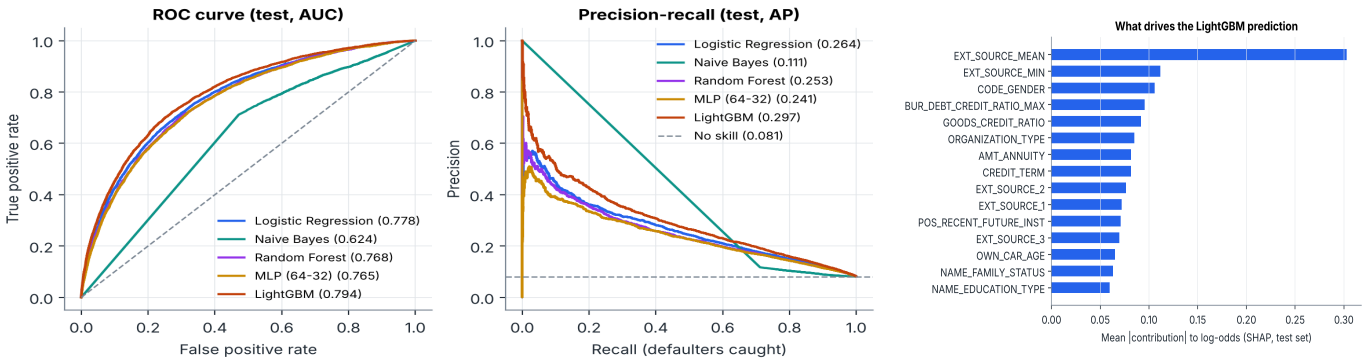


Figure 1. Left: ROC and precision-recall curves. Right: mean absolute SHAP contribution of the top 15 features in LightGBM.

LightGBM is the best model (test ROC-AUC 0.794, PR-AUC 0.297, 3.7 times the no-skill 0.081), ahead of logistic regression (0.778) and random forest (0.768). Naive Bayes is near chance: its independence assumption fails on hundreds of correlated

features. The cross-validation spread for the top four models is ±0.004 or less, so these gaps are real. **History matters.** The same LightGBM on the application table alone scores 0.770; the history features add 0.024. **The threshold matters more than the algorithm.** At the default 0.5 cut-off LightGBM shows 92.0% accuracy but catches only 5.7% of defaulters; at the tuned cut-off of 0.17 it catches 44%. **Resampling does not improve the model.** Class weights, under-sampling and SMOTE lower ROC-AUC by 0.0005 to 0.008; they only move the effective threshold, which tuning does for free. This contradicts the paper's finding that down-sampled data gives the best performance. **The paper's evaluation, reproduced.** The paper reports accuracy on down-sampled 50/50 data. On a 50/50 test set every one of our models beats its counterpart: LightGBM 72.1% (paper 57.5%), logistic regression 70.6% (69.3%), random forest 67.5% (63.5%), MLP 69.7% (69.2%), Naive Bayes 58.8% (52.1%).

K-means does not help. One LightGBM per cluster scores 0.792, 0.788, 0.784 and 0.782 AUC for k = 2 to 5, each below the single global model (0.794): splitting leaves each model fewer examples, and boosted trees already form their own segments. **What drives risk.** The external scores dominate (default rate falls from 23% in their lowest decile to 2% in the highest), followed by gender, the largest bureau debt/credit ratio, goods/credit, employer type, annuity and repayment speed. Without the external scores AUC is still 0.774. The model also holds up for the people the paper set out to help: 0.779 AUC for applicants missing two or more external scores (12% of test) and 0.785 for those with no credit-bureau record (14%). **Gender.** Dropping CODE_GENDER costs only 0.0015 AUC and narrows the share of men flagged as high risk from 19.2% to 15.9% (women: 11.0% to 10.5%). **As a policy:** reviewing the riskiest 10% of applicants catches 39% of defaulters and cuts the default rate among approved loans from 8.1% to 5.5%; reviewing 20% catches 58% and leaves 4.2%.

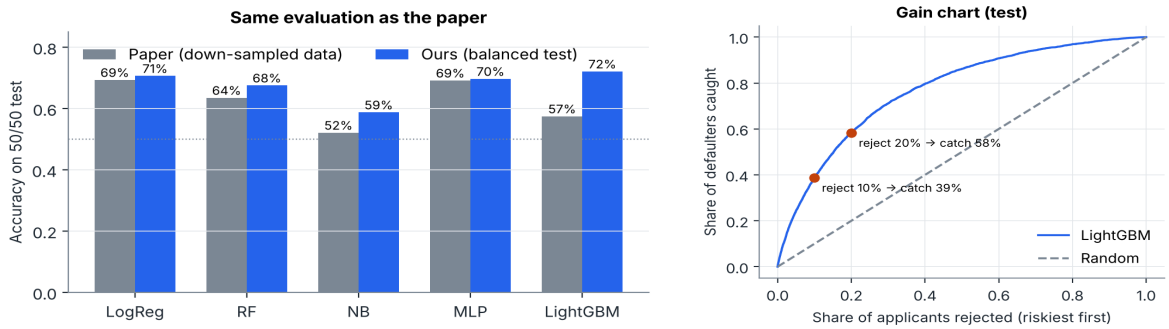


Figure 2. Left: accuracy on a 50/50 test set, paper vs ours. Right: share of defaulters caught when reviewing the riskiest applicants first.

5. Conclusions

Gradient-boosted trees on application data, affordability ratios and aggregated repayment history give the strongest and most stable model (ROC-AUC 0.794), in line with strong single models on this benchmark. History was the largest single improvement. On data this imbalanced, the metric and the threshold matter more than the algorithm, and resampling and clustering, both central to the reference paper, added nothing once evaluation was leak-free. The model stays accurate for applicants without bureau records or external scores. A deployable version should drop gender (it costs almost nothing), audit approval rates across groups, calibrate the probabilities, and set the threshold from the lender's cost of a default versus a lost customer rather than from F1.